RF Intelligence for Health: Classification of SmartBan Medical Signals in the overcrowded ISM frequency band

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Abstract— **Accurate classification of radio-frequency (RF) signals is essential for the development of reliable wearable health-monitoring systems. However, in dense spectral environments, such as ISM band, the identification of transmissions from medical devices remains a significant challenge. In this work, we present a novel framework for the automatic recognition and classification of RF emissions within Body Area Networks (BANs), with a specific focus on signals emitted by SmartBAN medical sensors. Our approach combines the generation of a synthetic dataset—comprising simulated signals in the ISM frequency band (2.4–2.48 GHz)—with the collection of real-world RF captures using Software Defined Radios (SDRs). These data are then used to train deep convolutional neural networks, specifically ResNet50 and ResNet18 backbones integrated with U-Net decoders enhanced by attention gates. We evaluate the performance of the proposed framework across multiple scenarios, including varying channel conditions, demonstrating its robustness and effectiveness.**

Keywords — RF Signals Classification, SDR, Images Segmentation, ISM band, Deep Learning

I. Introduction

Body Area Networks (BANs) are becoming essential in modern healthcare, enabling continuous and non-invasive monitoring of vital signs through wearable medical devices. These technologies are now part of everyday life, with smartwatches and fitness trackers as common examples.

However, in crowded RF environments as depicted in Figure 1, critical medical transmissions from BAN nodes can be masked by overlapping signals. This interference threatens the timely and reliable delivery of potentially life-saving data. As a result, there is a growing need for robust methods that can accurately detect and classify BAN-related signals in complex spectral conditions.

In this study, we focus on an emerging waveform standard for BANs. Although not yet widely deployed, this standard features a clear spectral signature and structured protocol. A key technical challenge is the extremely low power density of SmartBAN transmissions. In the 2.4–2.48 GHz ISM band, they often compete with higher-power signals from technologies like WLAN, ZigBee, and Bluetooth, which can easily overshadow them.

To address this issue, we propose Smart RF Signal Recognition, a deep learning framework designed to detect and classify SmartBAN medical waveforms. We begin by analysing the spectral behaviour of these transmissions in the presence of typical ISM-band interferers. Based on this analysis, we develop a lightweight neural network architecture using ResNet encoders and U-Net decoders, enhanced with attention mechanisms. The system is optimized for dynamic channels and mixed-signal environments.

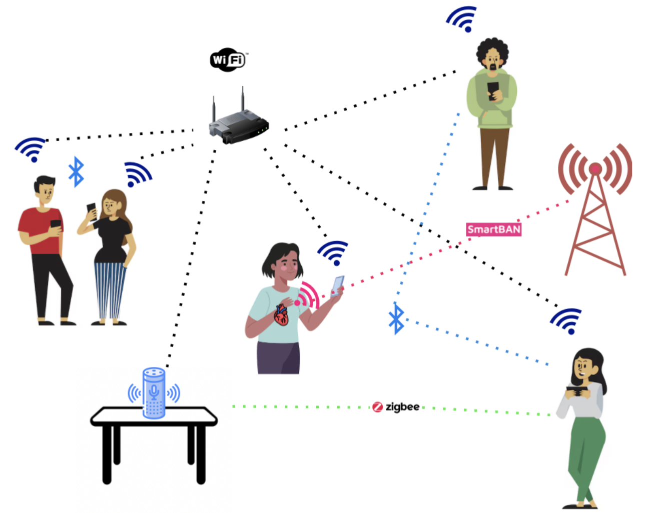

Fig. 1: Illustrated scenario of a crowded signals environment with used technicologies

Our approach combines synthetic simulations with real-world signal captures. Results show strong classification performance and robustness under conditions with additive white Gaussian noise (AWGN).

A. Background and state of the art

In recent years, AI-based signal detection systems have been increasingly applied to classify wireless transmissions in the crowded 2.4 GHz ISM band. Several studies have explored machine learning methods such as support vector machines (SVMs) [1] and convolutional neural networks (CNNs) [2] to recognize coexisting standards like Wi-Fi, ZigBee, and Bluetooth, even in overlapping transmission scenarios.

However, these approaches typically focus on signals with relatively high power and distinct spectral features. In contrast, SmartBAN medical waveforms—part of emerging Body Area Network (BAN) technologies—are characterized by extremely low power levels. As a result, they are more easily masked by stronger co-channel signals.

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To overcome this limitation, we propose a different strategy. Our method involves generating synthetic spectrograms that simulate mixed-technology environments, including ZigBee, WLAN, Bluetooth, and SmartBAN signals. Each signal within the spectrogram is individually labeled. These labeled spectrograms are then used to train a convolutional neural network capable of detecting and classifying BAN-related transmissions, even under adverse signal conditions.

B. *SmartBAN Standards*

In this subsection, we provide a brief overview of the latest SmartBAN standard, currently under development by the ETSI SmartBAN Technical Committee (TC). The goal is to offer minimal background information for the reader; for a complete and formal description, please refer to the official publications [3].

SmartBAN is a technical standard for Body Area Networks, designed to support ultra-low-power radio communication [4] with a simplified Medium Access Control (MAC) layer [4]. Although it has not yet been implemented in commercial devices, we recognize the relevance of the work carried out by the Technical Committee. For this reason, we include a simulation of what a SmartBAN transmission might look like when captured and visualized as a spectrogram.

A key feature of SmartBAN is the Smart Coordinator—a central device responsible for managing channel access among the network nodes (e.g., sensors monitoring physiological parameters). The coordinator also acts as a gateway between the BAN and external systems. To reflect this architecture, we modelled the signal as a high-power beacon packet (sent by the coordinator), followed by a series of lower-power data packets (transmitted by the nodes).

The modulation scheme and sampling rate used in the simulation follow the specifications provided by the Technical Committee. For parameters not explicitly defined in the documentation, we relied on empirical values drawn from existing literature (e.g., [5]).

II. Dataset generation

In this section, we highlight the main features of the generated signals and analyze their interaction and coexistence within the shared spectrum.

A. *Signal Types and Spectrogram Generation*

To realistically emulate wireless coexistence scenarios within the congested 2.4–2.48 GHz ISM band, we synthesized spectrogram data based on three widely used wireless standards: Wi-Fi (IEEE 802.11ax), Bluetooth (IEEE 802.15.1), and ZigBee (IEEE 802.15.4). These protocols were chosen to reflect a broad range of real-world applications, covering both personal and industrial use cases, and to capture the complexity typical of crowded RF environments.

Using MATLAB, we generated individual waveforms for each protocol, carefully adjusting parameters to match their specific transmission characteristics (summarized in Table I). To introduce realistic signal variability, each waveform was passed through a fading channel, modeled either as Rayleigh or Rician. We then applied additive white Gaussian noise (AWGN) to simulate signal-to-noise ratios (SNR) ranging from 0 to 30 dB, referenced to the peak power in the spectrogram.

Parameters	**BL**	**ZB**	**WLAN**	**SBAN**
Sample Rate	80 MHz	80 MHz	80 MHz	80MHz
Symbol Rate	1 MHz	4 MHz	20 MHz	1 MHz
Time Span	20 ms	20 ms	20 ms	20ms
Duration	(625 μs)**	4.2565 ms*	180 μs*	1,25 ms**
Idle Time	625 μs	(0 - 5μs)	20 μs	-

* Packet duration;
** Slot duration (BL may use 1, 3, or 5 slots)

TABLE I: Signal parameters used for dataset creation

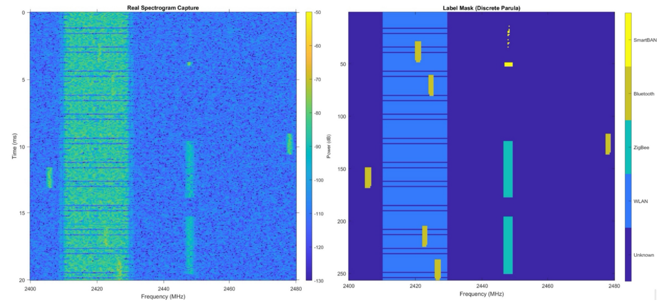


Fig. 2: Dataset spectrogram and labeled image of all signals at high SNR

To build the dataset, we created composite signals by randomly combining one to four individual transmissions, allowing overlaps in both time and frequency domains. Each component was scaled to its target SNR before mixing, ensuring realistic power dynamics and interference patterns. Finally, the composite waveforms were transformed into spectrograms using the short-time Fourier transform (STFT), producing the input images used for training and evaluation.

B. *Synthetic Dataset Adaptation for Training*

To enable supervised learning, we generated corresponding label masks by thresholding the spectrograms of the isolated signals. Each class was assigned a unique integer value (Fig. 2). In cases where signals overlapped in the time-frequency domain, we applied a deterministic priority scheme to resolve conflicts: Bluetooth > SmartBAN > ZigBee > Wi-Fi. This approach ensured consistent and unambiguous labeling.

All spectrogram images, along with their associated masks, were saved for training purposes. Using an NVIDIA GPU-accelerated pipeline, we generated approximately 30,000 labeled spectrogram samples.

In addition to the synthetic dataset, we collected real-world signal captures to validate and enhance the training process, as described in [6].

III. Spectrogram-based Semantic Segmentation with Deep Learning

After acquiring the dataset, we developed a flexible and fully automated MATLAB framework for semantic segmentation of spectrogram images. The goal was to identify and classify different types of radio signals in noisy environments.

The system processes RGB spectrogram tiles of size 256×256 pixels and assigns each pixel to one of five semantic classes: Unknown (noise/AWGN), WLAN, ZigBee, Bluetooth, or SmartBAN. These classes are encoded using a custom bitmask labeling scheme with integer values: 0, 16, 32, 64, and 128, respectively. This encoding ensures both efficient data representation and compatibility with MATLAB's pixel-level segmentation tools.

As part of the development, we tested multiple network architectures to optimize performance. All experimental results, configurations, and models are documented in the previously referenced GitHub repository [6].

A. Networks architecture

The proposed framework supports two state-of-the-art deep semantic segmentation models:

- **U-Net** (Fig. 3), a widely used encoder-decoder architecture featuring skip connections;
- **DeepLab v3+** (Fig. 4), which incorporates atrous spatial pyramid pooling (ASPP) to capture multiscale contextual information.

Both models can be configured with either a **ResNet-18** or **ResNet-50** backbone, offering a trade-off between network depth and computational cost.

A key innovation in our implementation is the optional integration of *additive attention gates* within the U-Net architecture. These modules are inserted into the skip connections between the encoder and decoder and are designed to enhance spatial focus during feature propagation. By generating soft attention maps—conditioned on both encoder (key) and decoder (query) features—the network learns to suppress irrelevant background activations and highlight class-relevant structures.

This mechanism is particularly beneficial in scenarios with significant background noise or overlapping transmissions, improving overall segmentation performance.

B. Dataset Management and Preprocessings

The input dataset is organized as pairs of spectrogram images (stored in `.png` format) and corresponding semantic masks (stored in `.mat` files). Each mask contains a single variable holding a 2D label matrix.

We used MATLAB's `imageDatastore` and `pixelLabelDatastore` functions to efficiently manage image-label pairs. These were then combined into a `pixelLabelImageDatastore` to enable streamlined batch processing.

The dataset is automatically split into training (70%), validation (10%), and test (20%) subsets using a reproducible random partition. This split preserves class distributions across the subsets, ensuring reliable evaluation and preventing data leakage.

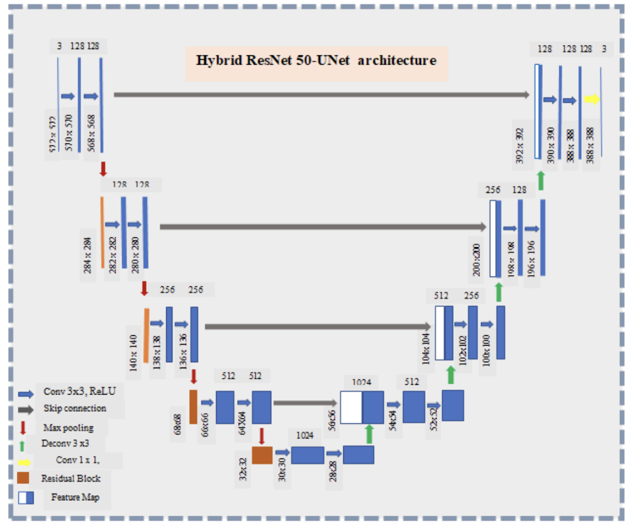
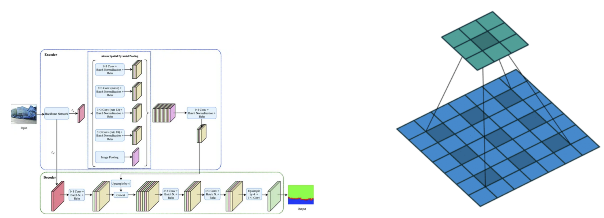


Fig. 3: Network architecture scheme [7]



(a) Architecture scheme (b) Atrous Convolution

Fig. 4: DeepLabv3+ architecture and convolution scheme [8]

C. Loss Function and Class Imbalance Handling

Given the naturally unbalanced distribution of signal types in real-world wireless environments, the training loss incorporates inverse class frequency weighting. Specifically, the pixel-wise cross-entropy loss is modulated by class-specific weights computed from the label histogram of the training set. This approach ensures that underrepresented classes (e.g., SmartBAN) contribute proportionally more to the loss, thereby reducing classifier bias toward dominant categories such as WLAN or Unknown.

The framework is also designed to support additional loss components, such as the Dice coefficient. This can be linearly combined with the cross-entropy loss to further improve boundary accuracy and better handle sparse class distributions. However, the Dice loss is disabled by default to maintain compatibility with the semantic segmentation training interface.

D. Training Strategy

Training is conducted using the Adam optimizer with a step-wise learning rate decay schedule. The initial learning rate is set to 8×10^{-4} and is reduced by a factor of 10 every 10 epochs. The default configuration includes 25 training epochs and a mini-batch size of 32.

Parameter	Value
Optimizer	Adam
Initial learning rate	8×10^{-4}
Learning rate decay	0.1 every 10 epochs
Total epochs	25
Mini-batch size	32

TABLE II: Training hyperparameters used in the segmentation pipeline

Networks	GlobalAccuracy	MeanAccuracy	MeanIoU	WeightedIoU	MeanBFScores
ResNet18	97,45%	NaN	NaN	95,84%	80.86%
ResNet50	96,72%	NaN	NaN	94.62%	72.93
ResNet18 A/N	99,29%	76,68%	74,97%	98,61%	92,19%
ResNet50 A/N	99,24%	75,55%	73,79%	98,51%	91,08%

A/N: with attention gates and noise;
NaN: Not a number

TABLE III: Networks evaluation parameters

Network	Prediction Mean (s)	Prediction Variance
ResNet18	0.2302	0.0031
ResNet18 (D)	0.2708	0.0158
ResNet18 (A/N)	0.3032	0.0144
ResNet50	1.7009	0.0109
ResNet50 (A)	1.8123	0.0236
ResNet50 (A/N)	1.7576	0.0103

A: Attention gate
A/N: Attention gates and Noise
D: Deep
A: Attention gate
* Tests conducted on a MacBook Pro with Apple M1 chip.

TABLE IV: Average prediction time and variance for each network

Validation is performed at regular intervals, and training progress is monitored using loss curves and per-class accuracy plots.

The framework is fully compatible with different computational backends, supporting: CPU-only execution, single-GPU training, multi-GPU parallelism, and automatic hardware selection.

E. Evaluation protocol

Once training is completed, the final model is evaluated on the hold-out test set using a batch-wise pixel-level inference function. The predicted segmentation maps are then compared against the ground truth masks using MATLAB's `evaluateSemanticSegmentation` function.

This evaluation yields a comprehensive set of metrics, including:

- **Pixel Accuracy**: providies a general measure of overall accuracy.
- **Mean Intersection-over-Union (mIoU)**: balances precision and recall, making it a robust measure for segmentation tasks.
- **Weighted F1-score**: reflects the balance between false positives and false negatives, especially useful when classes are imbalanced.
- **Class-wise confusion matrix**: highlights which classes are most often confused by the model, aiding in error analysis.

These metrics provide detailed insights into the model's performance across all signal types, particularly in distinguishing between structurally similar classes such as ZigBee and Bluetooth.

F. Network performance

The initial networks (ResNet50 and ResNet18) were trained on a synthetic, noise-free dataset and achieved satisfactory results in spectrogram recognition and signal classification. However, they showed limitations in accurately detecting signal boundaries, as reflected by a Boundary F1-score (BF-score) of approximately 80%. Additionally, their performance on real-world spectrograms was suboptimal, with noise often misclassified as valid signal activity.

To overcome these issues, later models were enhanced with attention gate mechanisms and trained on noise-augmented datasets. These improvements resulted in significantly better classification performance on real signal captures.

The results across the evaluated scenarios (using ResNet50) are summarized in Table III and Fig. 5.

IV. Experimental Results

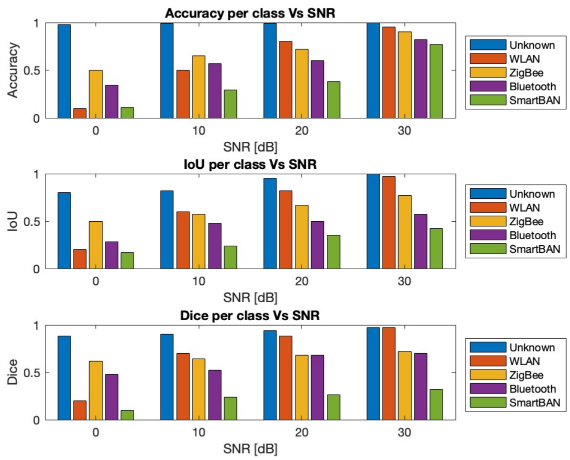
After training various neural networks on synthetic datasets, we tested their performance on real-world captures of the ISM frequency band. For this purpose, we employed multiple ADALM-PLUTO Software-Defined Radios (SDRs), which acted both as receivers for ambient signals and as transmitters to inject specific signal types into the environment.

In the following sections, we first describe the experimental setup and how we addressed hardware limitations. Then, we present the results, discussing the successes achieved as well as the challenges encountered with real-world data.

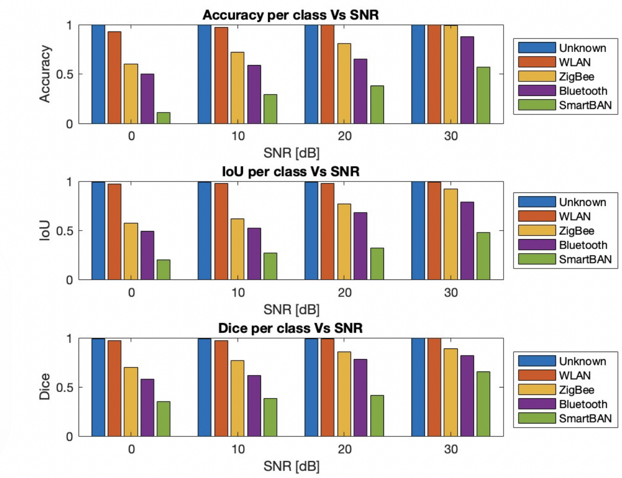
A. Experimental Setup

ADALM-PLUTO SDRs are widely used in scientific experiments due to their versatility and efficacy for signal analysis. However, their burst mode sampling rate is limited to 61.4 megasamples per second (MSps), which is insufficient for capturing the entire ISM band in a single acquisition. To overcome this, we adopted an approach inspired by the setup in [5] (detailed in Section III and illustrated in Figs. 6 and 7), which utilizes external synchronization from an independent board. This setup allows two simultaneous 40 MHz captures—centered at 2.42 GHz and 2.46 GHz respectively—to be aligned and combined through digital signal processing, effectively generating a continuous 80 MHz-wide spectrogram.

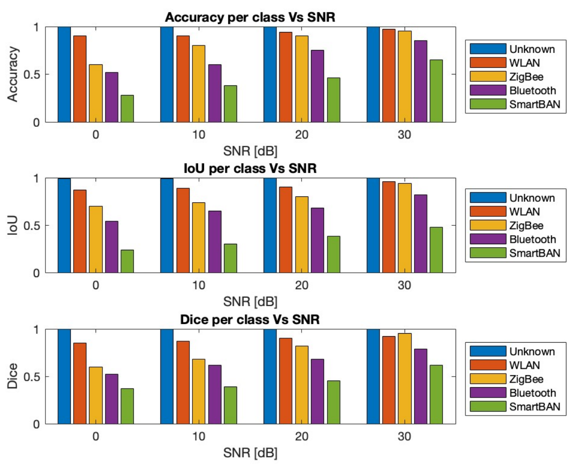
Using MATLAB's parallel computing tools, each PLUTO device's capture process was assigned to a separate thread,



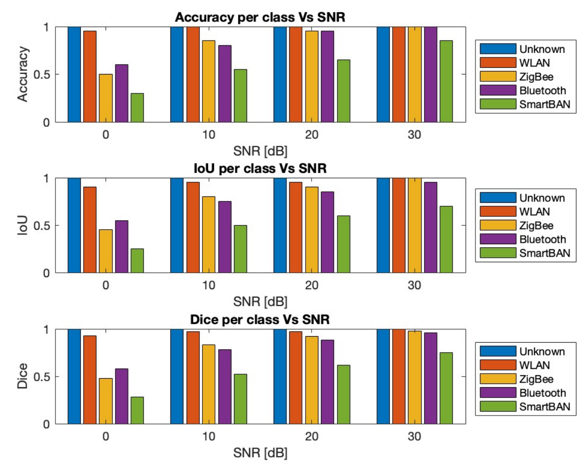
(a) ResNet 18



(b) ResNet 18 A/N



(c) ResNet 50



(d) ResNet 50 A/N

Fig. 5: Accuracy vs SNR outcomes in ResNet models under clean and noisy conditions

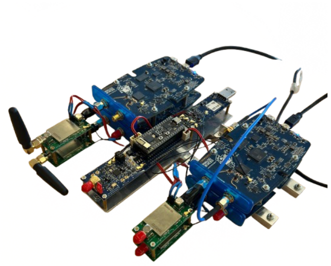


Fig. 6: Setup used for the receiver. The two PLUTO devices on the right are used for capturing, the left side top PLUTO is used together with the board in the middle to guarantee synchronization

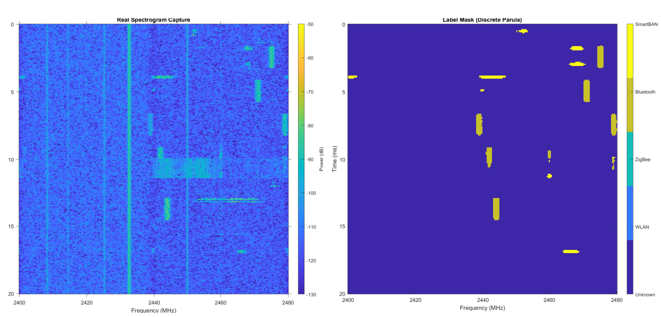


Fig. 7: Example of capture of Bluetooth packets originated from wireless earbuds. As it can be seen, unknown packets and small FM-modulated tones do not appear on the prediction, showing good resilience against noise (Prediction generated using ResNet50 with attention gates)

enabling concurrent acquisition and synchronization of the two frequency bands.

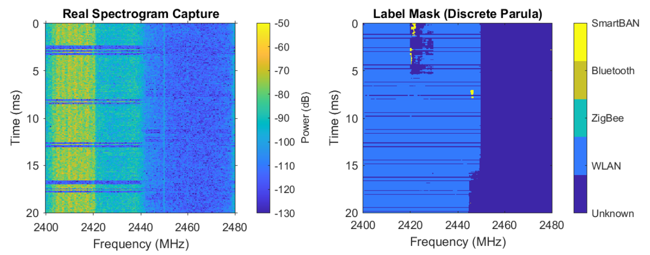


Fig. 8: The proposed system generally recognizing the signal

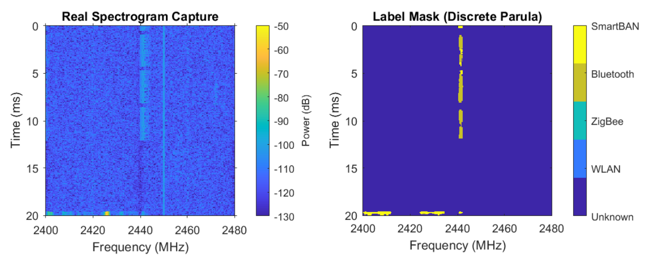


Fig. 9: The image show the incorrect prediction of the ZigBee signal. The difference in how the packets appear here compared to the synthetic image in Fig. 2 reinforce the hypothesis the channel is affecting the signal in unpredicted ways

B. Testing Environment

To test the described setup, MATLAB was used to generate waveform signals, which were then transmitted via a PLUTO device. However, because PLUTO devices cannot switch carrier frequencies rapidly enough to emulate Bluetooth Frequency Hopping Patterns, real-world Bluetooth signals (e.g., from Bluetooth earbuds) were captured by activating multiple Bluetooth devices simultaneously near the receiver.

All captures and tests were conducted indoors, primarily in large, sparsely occupied rooms such as study halls, to minimize uncontrolled interference.

C. Outcomes

The results demonstrate varying levels of success along with areas for improvement.

Incorporating interference into the training dataset and adding attention gates to the network architecture greatly enhanced resilience against the multiple noise sources present in the crowded ISM band. As shown in Fig. 8, the model successfully ignores various interfering signals and noise during prediction. Bluetooth packets are consistently identified with high precision, even when overlapping with other signals, with only minor miss-classifications occurring around smaller, unknown packets.

Due to the simulation approach for SmartBAN signals —modelled as small, low-power bursts—the network tends to exhibit a bias toward labeling faint energy spikes as SmartBAN rather than Unknown. While these results are promising, precise detection and classification remain exploratory, since actual SmartBAN devices are not yet commercially available. Improving these outcomes could involve increasing the diversity of interference signals in the training data.

Detection of WLAN and ZigBee signals proved more challenging. Although performance on synthetic images was strong, the network struggled to consistently and accurately identify wider-band signals in real captures (Fig. 9). In particular, confusion between ZigBee and Bluetooth was observed. This discrepancy, absent in synthetic tests, suggests that real-world channel effects may alter signal power density, leading to systematic underestimation of the occupied bandwidth and consequent misclassification as Bluetooth (Fig. 10).

Future work could focus on enhancing the realism of the training dataset by incorporating more diverse channel effects, thereby improving the model's robustness to real-world signal variations.

V. Conclusions

In conclusion, this work underscores the importance and potential impact of intelligent RF signal classification within medical Body Area Networks. While the results obtained are encouraging, further enhancements are necessary to improve the robustness and efficiency of the detection pipeline, especially in real-world environments characterized by high interference. Building on these findings, future efforts will focus on refining the framework by incorporating more realistic channel modelling to enhance dataset fidelity. Additionally, work will be directed towards optimizing the system for deployment on resource-constrained embedded platforms, with the goal of achieving real-time, on-device signal recognition. These advancements have the potential to enable smarter and more reliable wearable systems, capable of autonomous operation in complex and challenging RF environments.

References

[1] Simone Grimaldi, Aamir Mahmood, and Mikael Gidlund. *An SVM-Based Method for Classification of External Interference in Industrial Wireless Sensor and Actuator Networks*. URL: https://www.researchgate.net/publication/320136046_An_SVM-Based_Method_for_Classification_of_External_Interference_in_Industrial_Wireless_Sensor_and_Actuator_Networks.

[2] Mingju He et al. *Identification of ISM Band Signals Using Deep Learning*. URL: https://ieeexplore.ieee.org/abstract/document/9114911.

[3] *ETSI SmartBan Technical Committee web page*. URL: https://www.etsi.org/committee/1413-smartban.

[4] ETSI SmartBAN Technical Committee. *Smart Body Area Network (SmartBAN) Low Complexity Medium Access Control (MAC) for SmartBAN*. Technical Specification ETSI TS 103 325 V1.2.1. Accessed: 2025-07-18. ETSI, 2016. URL: https://www.etsi.org/deliver/etsi_ts/103300_103399/103325/01.02.01_60/ts_103325v010201p.pdf.

[5] Davide Scazzoli et al. "Experimental Comparison of UAV-Based RSSI and AoA Localization". In: *IEEE Sensors Letters* 8.1 (2024), pp. 1–4. DOI: 10.1109/LSENS.2023.3336375.

[6] Aragnetti Giacomo, Gallucci Nicola, and Malagrinò Matteo. *Git-Hub repository to access to source codes*. URL: https://github.com/nico-net/AICompetition.git.

[7] Fathia ABOUDI Peng, Cyrine DRISSI, and Tarek KRAIEM. *A Hybrid ResNet50-UNet Model for Ischemic Stroke brain Segmentation from MRI Images*. URL: http://paper.ijcsns.org/07_book/202211/20221167.pdf.

[8] Rajan Ghimire. *Semantic Segmentation Using DeepLabV3+ from scratch*. URL: https://medium.com/@r1j1nghimire/semantic-segmentation-using-deeplabv3-from-scratch-b1ff57a27be.